



AI Engineer Journey — 2026

Learn → Build → Improve → Repeat

I'm **Divyansh Gupta**, a Computer Engineering student documenting my journey to become a **skilled AI Engineer** — completely in public.

This repository is my **AI Engineering learning log, roadmap, experiments, projects, and progress tracker**.

I'm not trying to learn everything at once.

I'm learning one layer at a time, building with what I learn, sharing the process, and improving continuously.



The Goal

Become a skilled AI Engineer and build real-world AI systems.

My approach is simple:

```
LEARN
↓
BUILD
↓
BREAK
↓
UNDERSTAND
↓
IMPROVE
↓
SHARE
↓
REPEAT
```

This repository will evolve as I do.



My AI Engineer Roadmap — 2026

● Phase 1 — Foundation

Building the core technical foundation required for AI Engineering.

Python

- ☐ Python Basics
- ☐ Intermediate Python
- ☐ Advanced Python
- ☐ Object-Oriented Programming
- ☐ Python Projects

DSA

- ☐ Arrays
- ☐ Strings
- ☐ Linked Lists
- ☐ Stacks & Queues
- ☐ Trees
- ☐ Graphs
- ☐ Recursion
- ☐ Sorting & Searching
- ☐ 100–150 Problem Solving Practice

Git & GitHub

- ☐ Git Fundamentals
- ☐ Branching & Merging
- ☐ GitHub Workflow
- ☐ Pull Requests
- ☐ Collaboration
- ☐ Build a consistent public contribution history

APIs

- ☐ HTTP & REST APIs
- ☐ JSON
- ☐ API Authentication
- ☐ FastAPI
- ☐ Flask
- ☐ Build API-based projects

Cloud

- ☐ AWS Fundamentals
- ☐ EC2
- ☐ S3
- ☐ Lambda
- ☐ Basic Deployment

★ Foundation Project

Weather Application

API + Python + GitHub + Deployment

Phase 2 — Data Handling

Learning how to collect, clean, understand and visualize data.

SQL

- [] Queries
- [] Filtering
- [] Joins
- [] Subqueries
- [] Indexing
- [] Database Fundamentals

Pandas

- [] DataFrames
- [] Data Cleaning
- [] Grouping
- [] Filtering
- [] Data Transformation

NumPy

- [] Arrays
- [] Vector Operations
- [] Broadcasting
- [] Numerical Computing

Data Visualization

- [] Matplotlib
- [] Seaborn
- [] Data Exploration
- [] Visual Storytelling

Data Project

Clean → Analyze → Visualize → Share

Build a complete data-analysis project using a real dataset.

Phase 3 — Mathematics for AI

Understanding the mathematics behind machine learning instead of treating models like black boxes.

Statistics & Probability

- [] Probability Fundamentals
- [] Distributions

- ☐ Mean, Median & Variance
- ☐ Correlation
- ☐ Statistical Concepts

Linear Algebra

- ☐ Vectors
- ☐ Matrices
- ☐ Matrix Operations
- ☐ Eigenvalues & Eigenvectors

Calculus

- ☐ Derivatives
- ☐ Partial Derivatives
- ☐ Gradients
- ☐ Optimization

Main Focus

Understand the "WHY" behind ML models.

Especially:

Gradient Descent → Optimization → Model Behaviour

Phase 4 — Core AI / Machine Learning

Turning data and mathematics into intelligent systems.

Machine Learning

- ☐ Supervised Learning
- ☐ Unsupervised Learning
- ☐ Regression
- ☐ Classification
- ☐ Decision Trees
- ☐ Random Forest
- ☐ KNN
- ☐ Model Evaluation
- ☐ Feature Engineering
- ☐ Hyperparameter Tuning
- ☐ Scikit-learn

Deep Learning

- ☐ Neural Networks
- ☐ Forward Propagation
- ☐ Backpropagation

- ☐ CNNs
- ☐ RNNs
- ☐ Transformers
- ☐ PyTorch

★ ML Project

Build → Train → Evaluate → Improve → Deploy

The goal is not just to train models.

The goal is to understand the complete ML workflow.

💖 Phase 5 — Generative AI

Moving from traditional ML toward modern AI systems.

🧠 LLMs

- ☐ LLM Fundamentals
- ☐ OpenAI APIs
- ☐ Gemini APIs
- ☐ Claude APIs
- ☐ Prompt Engineering
- ☐ Structured Outputs
- ☐ Tool Calling

🔍 RAG — Retrieval-Augmented Generation

- ☐ Embeddings
- ☐ Semantic Search
- ☐ Vector Databases
- ☐ ChromaDB
- ☐ Document Processing
- ☐ Retrieval Pipeline
- ☐ RAG Evaluation

🔗 AI Application Frameworks

- ☐ LangChain
- ☐ LangGraph
- ☐ Tools
- ☐ Agents
- ☐ Workflows
- ☐ Automation

Final Projects

Build real-world AI applications such as:

- [] RAG Chatbot
- [] AI Agent
- [] AI Automation System
- [] Production-ready AI Application

Final Goal

Build → Deploy → Share → Improve

Every major project should ideally have a live demo, documentation and source code.



Progress Tracker

Phase	Area	Status
 1	Foundation	 In Progress
 2	Data Handling	 Upcoming
 3	Mathematics	 Upcoming
 4	Core AI / ML	 Upcoming
 5	Generative AI	 Upcoming

This tracker will be updated as I progress.



Current Focus: Python

I'm currently working through the **Python foundation**.

My Python learning isn't limited to watching tutorials.

I'm focusing on:

- Writing code
- Solving problems
- Understanding how Python actually works
- Running experiments
- Building mini-projects
- Debugging mistakes
- Exploring useful libraries
- Applying concepts to real problems

Python Learning Structure

```
Python
|
├── Fundamentals
├── Problem Solving
├── Data Structures
├── Functions
├── OOP
├── Modules & Packages
├── File Handling
├── APIs
├── Automation
├── Experiments
└── Mini Projects
```



My "Learn By Breaking Things" Approach

One thing I want to do differently:

I don't want to only learn what works.

I also want to understand:

Why does it work?

So alongside normal learning, I'll be experimenting with:

- Python edge cases
- Unexpected behaviour
- Performance comparisons
- Different approaches to the same problem
- Debugging challenges
- Library experiments
- Small automation ideas

If something breaks, that's part of the learning process.



Projects

This section will grow as I build.



Current / Upcoming

- [] Python Mini Projects

- [] Weather API Application
- [] Data Analysis Project
- [] Machine Learning Project
- [] Deep Learning Project
- [] RAG Application
- [] AI Agent
- [] Production AI System

Learning in Public

I'm documenting this journey across multiple platforms.

GitHub

Code, experiments, projects and progress.

LinkedIn

Daily learning logs, insights and lessons.

YouTube

Long-form learning, experiments, challenges and projects.

Instagram

Short-form updates and the journey behind the scenes.

The goal is simple:

Don't just tell people I'm learning AI. Show them.

My Daily Learning Philosophy

1 Hour → Learn
2 Hours → Build
1 Hour → DSA / Problem Solving

+

Consistency

↓

Real Projects

↓

Real Skills

I don't want passive learning.

Learn something → use it immediately.

Mistakes I'm Trying to Avoid

1. Passive Learning

Watching tutorials without building.

Solution:

Learn → Build immediately.

2. Learning Alone

Getting stuck for days without asking for help.

Solution:

Communities → Documentation → Discussions → Collaboration.

3. Applying Too Late

Waiting until I "know everything" before building or applying.

Solution:

Start building and applying early.

What You'll Find Here

As this journey progresses, this repository will contain:

-  Learning Notes
-  Source Code
-  Experiments
-  Problem Solving
-  Projects
-  Data Analysis
-  ML Models
-  Deep Learning
-  GenAI Applications
-  AI Agents
-  Deployments

Everything doesn't need to be perfect.

It needs to be **real**.

Why I'm Doing This

AI Engineering is a huge field.

Instead of trying to become an expert overnight, I'm focusing on becoming **1% better every day**.

This repository is a record of that process.

If you find something useful here, feel free to explore it, learn from it, or build on it.

And if you notice something I can improve, I'm always open to learning.

The Long-Term Vision

```
Python
↓
Data
↓
Mathematics
↓
Machine Learning
↓
Deep Learning
↓
Generative AI
↓
AI Engineering
↓
Real-World AI Systems
```

The destination isn't just learning AI.

The goal is to become someone capable of building useful AI systems from scratch.

★ Follow the Journey

If you're interested in watching an aspiring AI Engineer learn, build and improve in public, feel free to follow the journey.

Learn → Build → Improve → Repeat.